

# Materials Science Optimization Benchmark Dataset for High-dimensional, Multi-objective, Multi-fidelity Optimization of CrabNet Hyperparameters

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## Abstract

Benchmarks are crucial for driving progress in scientific disciplines. To be effective, benchmarks should closely mimic real-world tasks while being computationally efficient, allowing for accessibility and repeatability. Developing surrogate models that can be indistinguishable from the ground-truth observation within the explored dataset bounds dramatically reduces the computational burden of running benchmarks without sacrificing quality, but this requires a large amount of initial data. In the fields of materials science and chemistry, relevant optimization tasks can be challenging due to their complexity, which includes hierarchical, noisy, multi-fidelity, multi-objective, high-dimensional, and non-linearly correlated variables. Additionally, they may include mixed numerical and categorical variables that are subject to linear and non-linear constraints. Simulating or experimentally verifying such tasks can be difficult, which is why benchmarks are essential. This study aimed to overcome these challenges by generating 173,219 quasi-random hyperparameter combinations across 23 hyperparameters and using them to train CrabNet on the Matbench experimental band gap dataset (computational runtime: 387 RTX-2080-Ti GPU days). The results were stored in a free-tier shared MongoDB Atlas dataset, creating a regression dataset that maps

hyperparameter combinations to metrics such as MAE, RMSE, computational runtime, and model size. Heteroskedastic noise was incorporated into the regression dataset, and bad hyperparameter combinations were excluded; percentile ranks were computed within each group of identical parameter sets to capture heteroskedastic noise rather than assuming Gaussian noise. This approach can be applied to other benchmark datasets, bridging the gap between optimization benchmarks with low computational overhead and realistically complex, real-world optimization scenarios.

**Keywords:** Bayesian optimization, benchmark dataset, CrabNet, high-dimensional optimization, hyperparameter optimization, multi-fidelity

## 1 Background & Summary

Industry-relevant optimization tasks in the physical sciences are often “hierarchical, noisy, multi-fidelity [1–4], multi-objective [5–7], high-dimensional [8–11], and non-linearly correlated while exhibiting mixed numerical and categorical variables subject to linear [12] and non-linear constraints.” [12–14] Existing benchmark datasets and platforms for materials and molecular optimization [15–22], while very useful, typically are single-objective, single-fidelity, low-dimensional, and ignore or simplify the influence of noise. The growing population of self-driving laboratories [23, 24] and attention-based composition models that succeed CrabNet [25] therefore have few publicly available surrogates against which their adaptive-design components can be exercised under realistic, hierarchical, noisy conditions.

To address this gap, we generated a large hyperparameter response surface for the original CrabNet composition-only band gap regression model. A quasi-random Sobol design [2] of 65,536 hyperparameter combinations was specified over a constrained search space spanning 23 numerical and categorical hyperparameters; each combination was repeated up to five times, yielding 327,680 scheduled training runs. Of these, 173,219 completed successfully on RTX-2080-Ti GPUs (387 GPU-days,  $\sim 4,614$  CUDA-core years), producing 41,550 unique parameter sets with 1–5 successful repeats each. The resulting regression dataset records four observed outputs — mean absolute error (MAE), root-mean-square error (RMSE), GPU runtime, and on-disk model size — for every successful run, and is released together with a trained surrogate so that downstream users can evaluate optimization strategies without re-running the underlying CrabNet training.

The dataset and surrogate have already been re-used to benchmark algorithms in the BayBE library [26] and the Ax platform via an interactive HuggingFace Space<sup>1</sup> and in an Acceleration Consortium Kaggle competition<sup>2</sup>, demonstrating that the dataset described here can serve as a realistic yet computationally cheap testbed for the broader BO and SDL community.

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<sup>1</sup><https://huggingface.co/spaces/AccelerationConsortium/crabnet-hyperparameter>

<sup>2</sup><https://www.kaggle.com/datasets/acceleration-consortium/crabnet-optimization-challenge-dataset>

## 2 Methods

### 2.1 Hyperparameter search space and sampling design

The 23 hyperparameters span CrabNet’s encoder configuration (e.g., `N`, `d_model`, `heads`, `dim_feedforward`, `dropout`), training schedule (`epochs_step`, `lr`, `weight_decay`, `batch_size`, optimizer betas), data augmentation / scaling switches (`emb_scaler`, `pos_scaler`, `eps`, `fudge`, `alpha`), and categorical choices (`criterion`, `elem_prop`, `bias`, `out_hidden4`). Continuous parameters were given physically meaningful upper and lower bounds, with mutually-incompatible configurations excluded a priori so that all sampled combinations are admissible. Quasi-random samples were drawn using the Ax platform’s Sobol generator, which provides low-discrepancy coverage of the joint search space and avoids the structured artefacts of pure random sampling.

### 2.2 CrabNet training and execution environment

For each sampled combination, CrabNet v2.0.8 (<https://github.com/sparks-baird/CrabNet>) was trained on each of the five Matbench [15] folds of the `matbench_expt_gap` composition-only experimental band gap task (<https://matbench.materialsproject.org/>). Python orchestration in `scripts/crabnet_hyperparameter/crabnet_hyperparameter_submitit.py` dispatched jobs to SLURM via Submitit on the University of Utah’s Center for High-performance Computing (CHPC). To improve cluster utilisation and reduce latency, hyperparameter sets (including repeats) were shuffled and divided into batches, then submitted for asynchronous evaluation. Per-run results — the four observed metrics plus the full input hyperparameter vector and provenance information — were streamed to a free-tier shared MongoDB Atlas collection via the MongoDB Data API. Walltime and pre-emption limits caused  $\approx 47\%$  of scheduled runs not to complete; the trade-off was considered acceptable for the throughput gains and is reflected in the distribution of successful repeats per parameter set (Section 4).

### 2.3 Aggregation and percentile ranking

Successful runs were exported from MongoDB and aggregated to the machine-learning-ready table described in Section 3. For each repeated parameter set we additionally computed within-group percentile ranks of MAE, RMSE, and runtime using the “dense” ranking method with `pct=True` in `pandas.core.groupby.GroupBy.rank`. These percentile ranks provide a non-parametric handle on heteroskedastic noise: rather than imposing an a-priori Gaussian noise model parameterised by mean and standard deviation, downstream surrogate models can sample from the empirical rank distribution within a parameter cell, faithfully reproducing the true (often heavy-tailed) noise structure of the underlying training process. The rank-augmented table is the canonical input for the surrogate models distributed alongside the dataset.

### 3 Data Records

The dataset and accompanying trained surrogates are deposited on Zenodo with persistent identifier [10.5281/zenodo.7694268](https://doi.org/10.5281/zenodo.7694268)<sup>3</sup>. The deposition contains the files listed in Table 1. The primary data record is `sobol_regression.csv`; the remaining files are derived artefacts retained for reproducibility of the trained surrogate.

**Table 1** Files included in the Zenodo deposition (DOI [10.5281/zenodo.7694268](https://doi.org/10.5281/zenodo.7694268)).

| File                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>sobol_regression.csv</code>               | Primary data table. One row per successful CrabNet training run (173,219 rows, 35 columns): a unique <code>_id</code> , batch <code>session_id</code> , <code>timestamp</code> , the 23 input hyperparameters, the executing GPU <code>hardware</code> string, and the four observed outputs <code>model_size</code> , <code>runtime</code> , <code>mae</code> , <code>rmse</code> together with their within-group percentile-rank columns <code>runtime_rank</code> , <code>mae_rank</code> , <code>rmse_rank</code> . Column ordering and units are documented in the data-preparation notebook (see Section 6). |
| <code>model_metadata.json</code>                | Surrogate-model metadata: feature schema, target list, train/test fold definitions, and library/version information used to fit the surrogate from <code>sobol_regression.csv</code> .                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>surrogate_models.pkl</code>               | Pickled scikit-learn surrogate ensemble fit on the full table; the artefact backing the HuggingFace Space deployment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>cross_validation_models_{0--4}.pkl</code> | Five pickled surrogate models fit on the corresponding Matbench fold splits, used in the technical validation (Section 4).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### 4 Technical Validation

The dataset’s suitability as a noisy, multi-fidelity, multi-objective benchmark rests on three properties of the raw runs: a meaningful number of within-cell repeats, broad coverage of each output metric, and an empirical (rather than imposed) noise distribution that varies with the operating point. The histograms in Figures 1–5 demonstrate each property in turn.

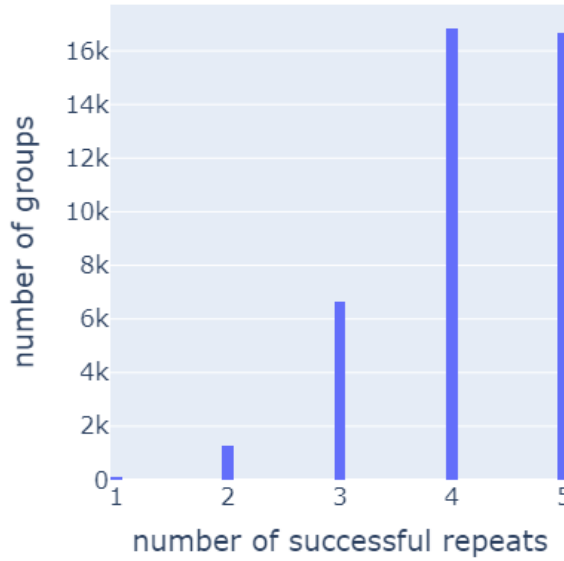
Figure 1 shows that, although the cap was five repeats per parameter set, the realised distribution of successful repeats is broad and right-skewed, with a mean of  $\approx 2.6$  successful repeats per set and 1–5 repeats observed across the 41,550 unique sets. This is the empirical signature of cluster pre-emption rather than a modelling assumption, and is precisely what makes the dataset a faithful multi-fidelity testbed: the same input can be queried at different effective fidelities (number of repeats), and the realised-fidelity distribution itself is heterogeneous across the search space.

Figures 2 and 3 show that MAE and RMSE both span more than two orders of magnitude across the explored search space, with long upper tails dominated by configurations that train unstable or under-parameterised models. This breadth ensures that optimization algorithms have to discriminate between qualitatively different regimes

<sup>3</sup><https://doi.org/10.5281/zenodo.7694268>

(well-trained, under-trained, divergent), not only between marginally different “good” configurations. Figure 4 (log-scaled) shows a similarly wide GPU-runtime distribution spanning roughly three orders of magnitude, which underpins the dataset’s multi-fidelity character: cheap configurations can be used as low-fidelity proxies for expensive ones in multi-fidelity acquisition functions (refs [1, 3, 4]). Figure 5 confirms that on-disk model size, the secondary objective, is also broadly distributed and non-trivially correlated with accuracy, providing a meaningful Pareto trade-off for multi-objective evaluation.

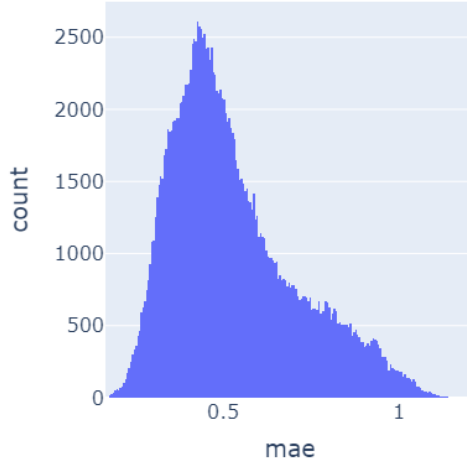
The five-fold cross-validation surrogate models distributed with the dataset (`cross_validation_models_{0--4}.pkl`, see Section 3) provide a held-out estimate of how faithfully the surrogate reproduces the underlying CrabNet response surface. Together, the repeat structure and the percentile-rank augmentation (Section 2.3) allow downstream users to draw heteroskedastic noise samples that match the empirical distribution within a parameter cell, rather than imposing an a-priori Gaussian noise model.



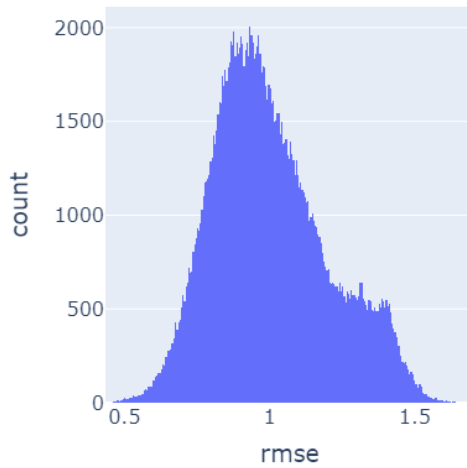
**Fig. 1** Histogram of number of parameter groups vs. number of successful repeats within a given group. The lowest number of successful repeats per parameter set is 1, with approximately 2.6 successful repeats on average.

## 5 Usage Notes

The primary table `sobol_regression.csv` can be loaded directly with `pandas.read_csv` for exploratory analysis. To use the dataset as a benchmark for adaptive design, we recommend the trained surrogate (`surrogate_models.pkl`)



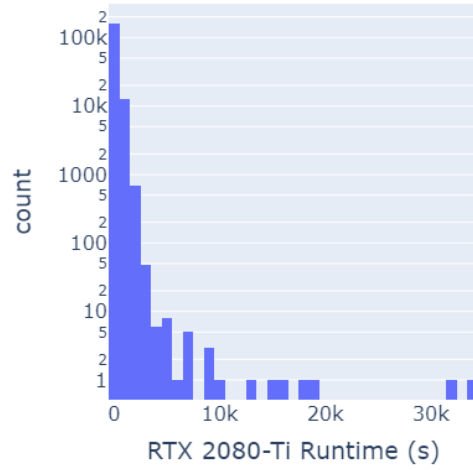
**Fig. 2** Histogram of number of training runs vs. mean absolute error using CrabNet on the Matbench experimental band gap task.



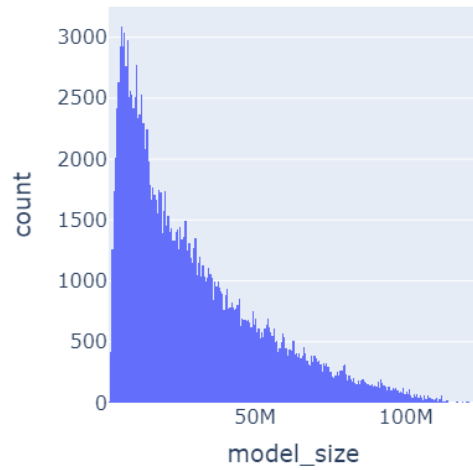
**Fig. 3** Histogram of number of training runs vs. root-mean-square error using CrabNet on the Matbench experimental band gap task.

instead: it exposes a callable that maps any in-bounds 23-dimensional hyperparameter vector to predicted MAE, RMSE, runtime and model size, together with an empirical noise sample drawn from the within-cell percentile-rank distribution. This avoids the cost of re-training CrabNet (387 GPU-days for the full table) while preserving the realistic, heteroskedastic noise structure that motivates the benchmark.

The same surrogate is also exposed as an interactive HuggingFace Space (see footnote in Section 1) and as a REST-callable model that has been used as the objective in BayBE and Ax benchmarking notebooks at <https://github.com/>



**Fig. 4** Histogram of number of training runs vs. GPU runtime on an RTX-2080-Ti using CrabNet on the Matbench experimental band gap task. The  $y$ -axis is log-scaled.



**Fig. 5** Histogram of number of training runs vs. model size using CrabNet on the Matbench experimental band gap task.

[AccelerationConsortium/baybe-multi-task-bo](https://www.kaggle.com/accelerationconsortium/baybe-multi-task-bo) and as the objective for an Acceleration Consortium Kaggle competition. These deployments are not part of the formal data record but are useful starting templates for downstream users wishing to benchmark a new acquisition function or optimisation library against the dataset.

## 6 Code availability

All code used to generate, aggregate, and surrogate-fit the dataset is openly available. The dispatch and aggregation pipeline lives in the `matsci-opt-benchmarks` repository under [https://github.com/sparks-baird/matsci-opt-benchmarks/tree/main/scripts/crabnet\\_hyperparameter](https://github.com/sparks-baird/matsci-opt-benchmarks/tree/main/scripts/crabnet_hyperparameter) and the data-preparation, surrogate-fitting and Zenodo-upload notebooks under [https://github.com/sparks-baird/matsci-opt-benchmarks/tree/main/notebooks/crabnet\\_hyperparameter](https://github.com/sparks-baird/matsci-opt-benchmarks/tree/main/notebooks/crabnet_hyperparameter). The exact source state used for the dataset is archived at Zenodo as release v0.2.1 with DOI 10.5281/zenodo.7694289<sup>4</sup>. The forked CrabNet implementation actually executed at training time is at <https://github.com/sparks-baird/CrabNet> (version 2.0.8).

## CRedit author statement

**Sterling G. Baird:** Project administration, Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization. **Jeet N. Parikh:** Methodology, Software, Formal Analysis, Data Curation, Writing – Original Draft, Writing – Review & Editing. **Taylor D. Sparks:** Supervision, Funding acquisition.

## Acknowledgments

**Funding:** This work was supported by the National Science Foundation, Division of Materials Research [Grant number DMR-1651668].

We thank Tupti Mohanty for work and discussion related to the future use-case of this dataset as a pseudo-materials benchmark. We acknowledge the University of Utah’s Center for High Performance Computing (CHPC) for providing computational resources. The code and manuscript associated with a benchmark dataset for particle packing simulations [13] served as a starting point for the code and manuscript for this work, for which there is shared language and document structure. We acknowledge OpenAI for providing free usage of their research tool, ChatGPT, which was used during the review and editing process.

## Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## References

- [1] Ghoreishi, S.F., Molkeri, A., Arróyave, R., Allaire, D., Srivastava, A.: Efficient use of multiple information sources in material design. *Acta Materialia* **180**, 260–271 (2019) <https://doi.org/10.1016/j.actamat.2019.09.009>

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<sup>4</sup><https://doi.org/10.5281/zenodo.7694289>



- [2] Kandasamy, K., Vysyaraju, K.R., Neiswanger, W., Paria, B., Collins, C.R., Schneider, J., Póczos, B., Xing, E.P.: Tuning hyperparameters without grad students: Scalable and robust bayesian optimisation with dragonfly. *Journal of Machine Learning Research* **21**(81), 1–40 (2020)
- [3] Sabanza-Gil, V., Barbano, R., Gutiérrez, D.P., Luterbacher, J.S., Hernández-Lobato, J.M., Schwaller, P., Roch, L.: Best practices for multi-fidelity Bayesian optimization in materials and molecular research. *Nature Computational Science* **5**(7), 572–581 (2025) <https://doi.org/10.1038/s43588-025-00768-8>
- [4] Gantzler, N., Deshwal, A., Doppa, J.R., Simon, C.M.: Multi-fidelity Bayesian optimization of covalent organic frameworks for xenon/krypton separations. *Digital Discovery* **2**, 1937–1956 (2023) <https://doi.org/10.1039/D3DD00117B>
- [5] Hanaoka, K.: Comparison of conceptually different multi-objective Bayesian optimization methods for material design problems. *Materials Today Communications* **31**, 103440 (2022) <https://doi.org/10.1016/j.mtcomm.2022.103440>
- [6] Häse, F., Roch, L.M., Aspuru-Guzik, A.: Chimera: Enabling hierarchy based multi-objective optimization for self-driving laboratories. *Chemical Science* **9**(39), 7642–7655 (2018) <https://doi.org/10.1039/C8SC02239A>
- [7] Ament, S., Daulton, S., Eriksson, D., Balandat, M., Bakshy, E.: Unexpected improvements to expected improvement for Bayesian optimization. In: *Advances in Neural Information Processing Systems 36 (NeurIPS)* (2023)
- [8] Baird, S.G., Liu, M., Sparks, T.D.: High-dimensional Bayesian optimization of 23 hyperparameters over 100 iterations for an attention-based network to predict materials property: A case study on CrabNet using Ax platform and SAASBO. *Computational Materials Science* **211**, 111505 (2022) <https://doi.org/10.1016/j.commatsci.2022.111505>
- [9] Eriksson, D., Jankowiak, M.: High-dimensional Bayesian optimization with sparse axis-aligned subspaces. In: *Proceedings of the 37th Conference on Uncertainty in Artificial Intelligence (UAI). Proceedings of Machine Learning Research*, vol. 161 (2021). <https://doi.org/10.48550/arXiv.2103.00349>
- [10] Hvarfner, C., Hellsten, E.O., Nardi, L.: Vanilla Bayesian optimization performs great in high dimensions. In: *Proceedings of the 41st International Conference on Machine Learning (ICML). Proceedings of Machine Learning Research*, vol. 235, pp. 20793–20817 (2024)
- [11] Papenmeier, L., Nardi, L., Poloczek, M.: Bounce: Reliable high-dimensional Bayesian optimization for combinatorial and mixed spaces. In: *Advances in Neural Information Processing Systems 36 (NeurIPS)* (2023)
- [12] Baird, S.G., Hall, J.R., Sparks, T.D.: Compactness matters: Improving Bayesian

- optimization efficiency of materials formulations through invariant search spaces. *Computational Materials Science* **224**, 112134 (2023) <https://doi.org/10.1016/j.commatsci.2023.112134> . Supersedes ChemRxiv preprints doi:10.26434/chemrxiv-2022-nz2w8-v2 and doi:10.26434/chemrxiv-2022-nz2w8-v3.
- [13] Baird, S.G., Issa, R., Sparks, T.D.: Materials science optimization benchmark dataset for multi-objective, multi-fidelity optimization of hard-sphere packing simulations. *Data in Brief* **50**, 109487 (2023) <https://doi.org/10.1016/j.dib.2023.109487> . Supersedes ChemRxiv preprint doi:10.26434/chemrxiv-2023-fjjk7.
  - [14] Hickman, R.J., Tom, G., Zou, Y., Aldeghi, M., Aspuru-Guzik, A.: Anubis: Bayesian optimization with unknown feasibility constraints for scientific experimentation. *Digital Discovery* **4**, 2104–2122 (2025) <https://doi.org/10.1039/D5DD00018A>
  - [15] Dunn, A., Wang, Q., Ganose, A., Dopp, D., Jain, A.: Benchmarking materials property prediction methods: The Matbench test set and Automatminer reference algorithm. *npj Computational Materials* **6**(1), 138 (2020) <https://doi.org/10.1038/s41524-020-00406-3>
  - [16] De Breuck, P.-P., Evans, M.L., Rignanese, G.-M.: Robust model benchmarking and bias-imbalance in data-driven materials science: A case study on MODNet. *Journal of Physics: Condensed Matter* **33**(40), 404002 (2021) <https://doi.org/10.1088/1361-648X/ac1280>
  - [17] Choudhary, K., Wines, D., Li, K., Garrity, K.F., Gupta, V., Romero, A.H., Krogel, J.T., Saritas, K., Fuhr, A., Ganesh, P., Kent, P.R.C., Yan, K., Lin, Y., Ji, S., Blaiszik, B., Reiser, P., Friederich, P., Agrawal, A., Tiwary, P., Beyerle, E., Minch, P., Rhone, T.D., Takeuchi, I., Wexler, R.B., Mannodi-Kanakkithodi, A., Ertekin, E., Mishra, A., Mathew, N., Wood, M., Rohskopf, A.D., Hattrick-Simpers, J., Wang, S.-H., Achenie, L.E.K., Xin, H., Williams, M., Biacchi, A.J., Tavazza, F.: JARVIS-leaderboard: A large scale benchmark of materials design methods. *npj Computational Materials* **10**, 93 (2024) <https://doi.org/10.1038/s41524-024-01259-w>
  - [18] Wang, A.E., Liang, H., McDannald, A., Takeuchi, I., Kusne, A.G.: Benchmarking active learning strategies for materials optimization and discovery. *Oxford Open Materials Science* **2**(1), 006 (2022) <https://doi.org/10.1093/oxfmat/itac006>
  - [19] Liang, Q., Gongora, A.E., Ren, Z., Tiihonen, A., Liu, Z., Sun, S., Deneault, J.R., Bash, D., Mekki-Berrada, F., Khan, S.A., Hippalgaonkar, K., Maruyama, B., Brown, K.A., Fisher III, J., Buonassisi, T.: Benchmarking the performance of Bayesian optimization across multiple experimental materials science domains. *npj Computational Materials* **7**(1), 188 (2021) <https://doi.org/10.1038/s41524-021-00656-9>
  - [20] Henderson, A.N., Kauwe, S.K., Sparks, T.D.: Benchmark datasets incorporating

- diverse tasks, sample sizes, material systems, and data heterogeneity for materials informatics. *Data in Brief* **37**, 107262 (2021) <https://doi.org/10.1016/j.dib.2021.107262>
- [21] Häse, F., Aldeghi, M., Hickman, R.J., Roch, L.M., Christensen, M., Liles, E., Hein, J.E., Aspuru-Guzik, A.: Olympus: A benchmarking framework for noisy optimization and experiment planning. *Machine Learning: Science and Technology* **2**(3), 035021 (2021) <https://doi.org/10.1088/2632-2153/abedc8>
  - [22] Hickman, R.J., Sim, M., Pablo-García, S., Tom, G., Woolhouse, I., Hao, H., Bao, Z., Bannigan, P., Allen, C.J., Aldeghi, M., Aspuru-Guzik, A.: Atlas: A brain for self-driving laboratories. *Digital Discovery* (2025) <https://doi.org/10.1039/D4DD000115J>
  - [23] Tom, G., Schmid, S.P., Baird, S.G., Cao, Y., Darvish, K., Hao, H., Lo, S., Pablo-García, S., Rajaonson, E.M., Skreta, M., Yoshikawa, N., Corapi, S., Akkoc, G.D., Strieth-Kalthoff, F., Seifrid, M., Aspuru-Guzik, A.: Self-driving laboratories for chemistry and materials science. *Chemical Reviews* **124**(16), 9633–9732 (2024) <https://doi.org/10.1021/acs.chemrev.4c00055>
  - [24] Abolhasani, M., Kumacheva, E.: The rise of self-driving labs in chemical and materials sciences. *Nature Synthesis* **2**, 483–492 (2023) <https://doi.org/10.1038/s44160-022-00231-0>
  - [25] Madani, M., Lacivita, V., Shin, Y., Tarakanova, A.: Accelerating materials property prediction via a hybrid transformer graph framework that leverages four body interactions. *npj Computational Materials* **11** (2025) <https://doi.org/10.1038/s41524-024-01472-7>
  - [26] Fitzner, M., Šošić, A., Hopp, A.V., Müller, M., Rihana, R., Hrovatin, K., Liebig, F., Winkel, M., Halter, W., Brandenburg, J.G.: BayBE: A Bayesian back end for experimental planning in the low-to-no-data regime. *Digital Discovery* (2025) <https://doi.org/10.1039/D5DD00050E>